

TAKAMATSU, Japan

WMO: 478910

Lat: 34.3183N

Lon: 134.0542E

Elev: 35

StdP: 14.68

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 31.4	(c) 33.1	(d) 16.3	(e) 12.6	(f) 38.8	(g) 18.8	(h) 14.1	(i) 40.2	(j) 21.3	(k) 44.1	(l) 19.4	(m) 43.0	(n) 5.1	(o) 220	(p) 0.380

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 12.9	(c) 94.2	(d) 77.5	(e) 92.4	(f) 77.2	(g) 90.5	(h) 76.9	(i) 79.5	(j) 89.3	(k) 78.8	(l) 88.3	(m) 78.1	(n) 87.2	(o) 8.8	(p) 270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 77.0	(b) 141.0	(c) 84.0	(d) 76.2	(e) 137.0	(f) 83.7	(g) 75.4	(h) 133.2	(i) 83.4	(j) 43.1	(k) 89.5	(l) 42.3	(m) 88.4	(n) 41.6	(o) 87.5	(p) 82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 17.7	(b) 15.2	(c) 13.2	(e) 28.4	(f) 97.4	(g) 1.8	(h) 1.5	(i) 27.2	(j) 98.4	(k) 26.1	(l) 99.3	(m) 25.1	(n) 100.1	(o) 23.9	(p) 101.2		
			DB	WB	25.0	80.8	2.0	0.9	23.6	81.5	22.4	82.0	21.3	82.6	19.9	83.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	62.4	42.3	43.2	49.0	58.7	68.0	74.4	82.2	84.2	76.9	66.7	55.8	46.6
	DBStd	15.22	3.69	4.41	5.19	5.60	4.22	4.15	4.01	3.33	4.92	5.07	5.38	4.55
	HDD50	658	240	196	83	5	0	0	0	0	0	0	9	125
	HDD65	2913	703	610	495	197	20	0	0	0	0	39	279	570
	CDD50	5199	1	6	53	266	557	732	998	1060	807	516	183	20
	CDD65	1980	0	0	0	9	112	282	533	595	357	90	2	0
	CDH74	18762	0	0	0	54	559	1736	5875	7291	2926	320	1	0
	CDH80	7186	0	0	0	5	75	421	2453	3331	864	37	0	0
(14) Wind	WSAvg	5.5	6.6	6.0	6.0	5.7	5.4	5.0	5.1	5.4	5.0	4.8	5.0	6.1
(15) Precipitation	PrecAvg	45.6	1.5	1.8	3.1	2.8	4.1	6.1	6.1	4.4	6.8	4.7	2.2	1.9
	PrecMax	63.8	3.9	3.4	7.8	6.7	11.9	13.7	13.8	17.0	20.8	16.4	5.6	4.4
	PrecMin	30.3	0.1	0.3	1.3	1.1	0.4	0.9	0.6	0.5	1.1	0.8	0.6	0.2
	PrecStd	10.4	0.9	0.9	1.5	1.1	2.4	3.3	3.7	3.9	5.0	3.9	1.3	1.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.4	61.8	69.3	79.3	85.5	91.4	95.9	96.9	92.9	83.9	72.3	62.7
		MCWB	48.4	52.7	54.4	59.6	64.9	75.3	77.5	77.8	77.1	72.2	62.6	55.5
	2%	DB	53.9	57.0	64.8	74.9	82.0	87.2	93.7	94.9	89.7	80.1	69.6	59.6
		MCWB	45.9	48.9	52.8	59.0	63.5	72.5	77.3	77.9	76.1	69.3	61.2	52.5
	5%	DB	51.7	54.3	61.8	71.9	79.5	84.5	91.9	93.1	87.2	77.3	67.5	57.3
		MCWB	44.3	46.8	51.5	57.8	63.0	70.7	77.2	77.4	74.8	67.2	59.9	50.2
	10%	DB	49.7	51.5	58.7	68.9	77.2	82.1	90.0	91.3	84.9	75.1	65.2	55.0
		MCWB	43.0	44.1	49.5	56.6	62.4	69.8	77.0	77.3	73.5	65.7	58.2	48.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.3	55.5	58.8	66.7	71.4	77.2	80.3	80.7	78.9	74.8	66.3	58.3
		MCDB	55.0	59.9	64.0	71.5	77.3	86.8	90.5	90.9	88.1	80.7	68.9	60.8
	2%	WB	47.7	51.2	55.7	63.2	69.0	75.5	79.2	79.8	77.8	72.0	63.7	54.2
		MCDB	51.6	55.4	61.0	69.4	74.9	83.4	89.1	89.7	86.2	77.8	67.5	58.0
	5%	WB	45.8	48.1	53.4	61.0	67.0	74.3	78.5	79.1	76.8	69.6	61.5	51.6
		MCDB	50.3	52.2	59.2	67.7	73.7	81.2	88.1	88.6	84.5	75.0	66.0	55.8
	10%	WB	43.8	45.6	50.9	59.1	65.4	72.7	77.8	78.5	75.7	67.4	59.2	49.3
		MCDB	48.8	50.3	57.5	66.0	72.8	79.0	86.9	87.7	82.6	73.2	64.2	54.0
(35) Mean Daily Temperature Range	MDBR		12.3	13.4	15.1	16.3	15.9	12.6	12.4	12.9	12.2	13.2	13.5	12.7
	5% DB	MCDBR	15.4	17.4	19.9	20.8	20.0	16.1	15.0	15.0	14.2	14.8	14.9	15.4
		MCWBR	10.5	11.2	11.7	9.5	8.2	5.8	4.5	4.7	5.5	7.5	9.3	10.5
	5% WB	MCDBR	14.4	14.5	16.3	15.8	14.5	12.8	13.7	13.6	12.2	11.7	12.7	13.5
		MCWBR	11.2	11.1	11.8	9.5	8.2	5.5	4.7	4.8	5.2	7.4	9.3	11.0
(40) Clear-Sky Solar Irradiance	taub		0.399	0.445	0.505	0.535	0.539	0.558	0.558	0.531	0.491	0.449	0.420	0.393
	taud		2.166	2.058	1.913	1.892	1.952	1.980	2.032	2.096	2.178	2.210	2.196	2.199
	Ebn at Noon		247	248	243	243	244	239	238	242	246	245	239	241
	Edh at Noon		37	46	58	62	59	57	54	50	44	39	36	34
(44) All-Sky Solar Radiation	RadAvg		785	974	1282	1544	1672	1494	1617	1665	1298	1071	815	696
	RadStd		54	75	93	141	159	144	206	163	125	106	69	44

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page